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SYLLABUS FOR MATH 383, SPRING 2026

Instructor: Xinyun Liu
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Office Hours: Friday 10-11 am, 13-14 pm, Chapman 450
TA Office Hours: McClinton, Gabriel Liam, email: glmcclin@unc.edu. Please reach out to our TA if you have any problems with the submission or grading of the homework.

Course Meeting Time & Place: MWF 9:05 AM-9:55 AM, Phillips Hall, 332

1. Course Information

Prerequisites: You must have earned a passing grade in MATH233 (or an equivalent) to register for this class. We suggest reviewing differentiation rules and integration techniques from single variable calculus.

Textbook: *Differential Equations: Computing and Modeling*, Sixth Edition, by Edward, Penney and Calvis.

Homework: Homework assignments will be handed out *via* Gradescope on Canvas.

Canvas: All handouts, solutions and other class material will be posted on Canvas, which can be accessed by going to: https://edtech.unc.edu/service/canvas/.

Course Content and Structure: Differential equations are equations involving relationships between functions and their derivatives. Differential equations are indispensable tools in describing physical processes and dynamical systems, and are vital in the study of the physical sciences. Differential equations are also of mathematical interest in their own right.

You are strongly encouraged to ask questions throughout this course in class, in office hours, and to each other. The more you are engaged in the material, the more beneficial this class will be. Group idea is highly recommended. Throughout the term, you should be reading along in the textbook, reviewing your notes, and working problems beyond what are assigned in the homework to get a better grasp of the material and to be better prepared for both lectures and the exams.

2. Class Expectations

Attendance: Attending class is necessary (but not sufficient) to succeed in this class. We will begin class promptly to use our time as effectively as possible, thus you should be on time and prepared to start class. Leaving class early will be counted as an absense. If you need to leave early, let me know in advance. You are only allowed two unexcused absences.

Electronic Devices: All cell phones and electronics, beside those needed for class, should be put away/on silent.

Honor Code Statement: Each student is expected to abide by the Honor Code and the Student Code of Conduct. See http://honor.unc.edu.

In this class, all exams must be done individually and are closed book and closed notes. It is an instance of cheating to give or receive help on an exam, except from the instructor. On homework assignments, students are encouraged to work together in pairs or small groups, provided that all participants are contributing and the collaboration benefits the learning of all involved. Simply copying or trading answers is an instance of cheating.

[1] Updated January 6, 2026

3. Assignments and Grading

Homework: Homework will primarily be assigned and submitted online through Gradescope in Canvas system. It is **your responsibility** to keep up with posted assignments and their due dates.

In Class Worksheets: I aim to give students the opportunity to work with the content presented in class before ever leaving class, so they can ask the questions that normally arise when they first open their homework. These worksheets are also how I will take attendance the majority of the time. Worksheets will be handed out on Gradescope prior to class and collected (submitted to Gradescope after class). Grading for worksheets will adhere to the following format: 10 points for correct answers + relevant work visible, 9 points for almost correct work but minor errors, 5 points for minimal effort, 1 point for no appreciable effort or "answers" without any work shown. (to clarify, you simply write down a correct answer to the problem with no work shown), 0 points if not turned in.

Exams: There will be two middle exams, lasting the entire class period.

Exam 1: Friday, February 20, 2026
Exam 2: Wednesday, April 1, 2026

There will be no make-up exams, except for religious and unversity-sponsored exceptions. You will not need a blue book or scantron.

Comprehensive Final Exam: will be on 12:00 PM - 3:00 PM, Wednesday, April 29, 2026, held in the regular assigned meeting room, Phillips Hall, 332, unless otherwise notified. The final exam is given in compliance with UNC's final exam regulations and calendar. In order to take a make-up exam after this date, you must have an official examination excuse, signed by a Dean or authorized agent of the Dean. You must bring this excuse and a picture ID to the make-up exam.

Late Work: No late tests or make-up tests will be given. Students who need to miss a test for a UNC athletic team event, UNC academic field trip, or religious holiday can take the test in absentia or in advance with at least a week of advance notice and written documentation. Exceptions will be made only in extreme circumstances with intervention from the Dean of Students office. Late written homework will not be accepted.

Grade Calculation:

Written Homework	10%
In Class Worksheets	10%
Exam 1	25%
Exam 2	25%
Final Exam	30%

Course Grade:

92-100	A	77-79	C+
90-92	A-	73-76	C
87-89	B+	70-72	C-
83-87	B	67-69	D+
80-82	B-	60-66	D
		0-59	F

There are no grades of D- or A+.

Week	Dates	Sections	Topics
1	01/07	1.1	Introduction to Differential Equations
	01/09	1.2	Integrals as General and Particular Solutions
	01/12	1.2	Existence and Uniqueness of Solutions
2	01/14	1.4	Separable Equations
	01/16	1.5	Linear First Order Equations 1
	01/19	No class	*Holiday (MLK Jr.)*
3	01/21	1.5	Linear First Order Equations 2
	01/23	1.6	Substitution Methods and Exact Equations 1
	01/26	1.6	Substitution Methods and Exact Equations 2
4	01/28	1.6	Substitution Methods and Exact Equations 3
	01/30	2.1	Population Model/Logistic Eqns 1
	02/02	2.1	Population Model/Logistic Eqns 2
5	02/04	2.2	Equilibrium Solutions & Stability
	02/06	3.1	Introduction to Second Order Equations
	02/09	No class	*Well-being Day*
6	02/11	3.1	Wronskian and Abel's theory
	02/13	3.2	General Linear Equations
	02/16	3.3	Homogeneous Constant Coefficient Equations 1
7	02/18	Review	
	02/20	**Exam 1**	Sections: 1.1, 1.2, 1.4, 1.5, 1.6, 2.2
	02/23	3.3	Homogeneous Constant Coefficient Equations 2
8	02/25	3.3	High Order Linear Homogeneous Equations
	02/27	3.4	Mechanical Vibrations
	03/02	3.4	Damped Vibrations
9	03/04	3.5	Non-Homogeneous Equations: Undetermined Coefficient 1
	03/06	3.5	Non-Homogeneous Equations: Undetermined Coefficient 2
	03/09	3.5	Non-Homogeneous Equations: Variation of Parameter 1
10	03/11	3.5	Non-Homogeneous Equations: Variation of Parameter 2
	03/13	3.8	Boundary Value Problems and Eigenvalues 1
	03/16	No class	*Spring Break*
11	03/18	No class	*Spring Break*
	03/20	No class	*Spring Break*
	03/23	3.8	Boundary Value Problems and Eigenvalues 2
12	03/25		Linear Algebra 1
	03/27		Linear Algebra 2
	03/30		Review
13	04/01	**Exam 2**	Section: 3.1, 3.2, 3.3, 3.5, 3.8
	04/03	5.1	Linear systems of Differential Equations
	04/06	5.2	Eigenvalue Problems for Homogeneous Systems (Real Distinct)
14	04/08	5.2	Eigenvalue Problems for Homogeneous Systems (Real Repeated) 1
	04/10	5.2	Eigenvalue Problems for Homogeneous Systems (Real Repeated) 2
	04/13	5.2	Eigenvalue Problems for Homogeneous Systems (Complex) 1
15	04/15	5.2	Eigenvalue Problems for Homogeneous Systems (Complex) 2
	04/17	No class	Spring Holiday
	04/20	5.4	Second Order Systems - Mass Springs and Eigenvalues
16	04/22	Review1	
	04/24	Review2	
	04/27	Review3	
17	04/29	**FINAL**	12:00 PM - 3:00 PM, Wednesday, April 29

TABLE 1. Approximate Lecture Schedule.